MID-SEMESTER RESEARCH PROJECT MILESTONE: ONE-PAGE SUMMARY

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Title: Rule-Constrained Reinforcement Learning for Ambiguity-Aware IT Incident Triage

Problem and research objective

Problem. When an IT incident is reported, support staff must act before they know the cause, and the first evidence is often incomplete or contradictory. They can check further, try a likely fix, or pass the incident to another technician. Current AI tools give one answer per ticket. They cannot ask for more evidence or signal that they are unsure [1].

Objective. To build and test an AI agent that learns when to check further, when to recommend a fix that the rules allow, and when to hand the incident to a human technician.

What exists, the gap and my contribution

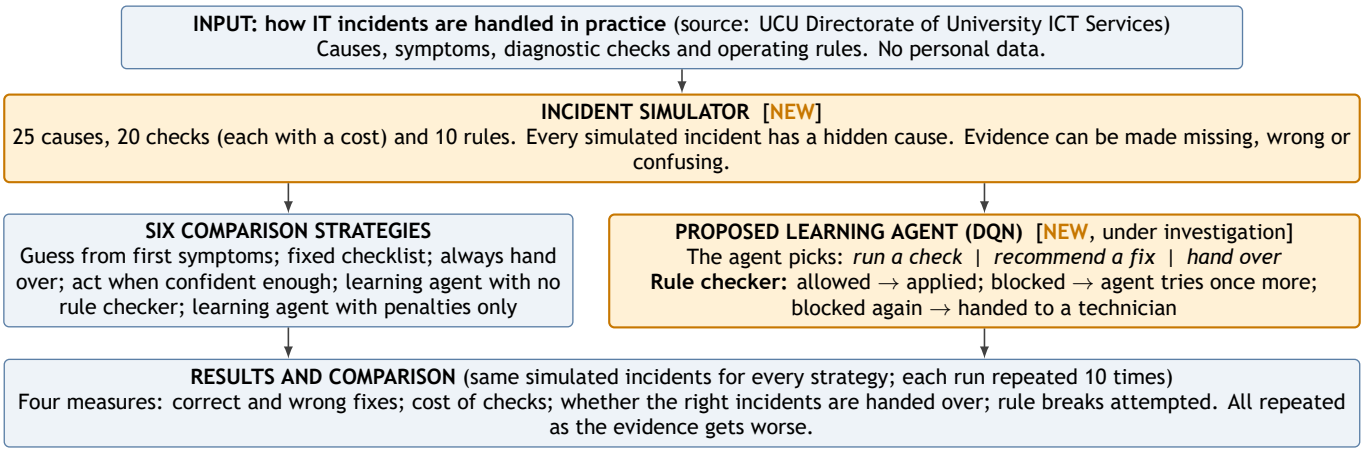
What exists.

- AI tools sort tickets using the report alone, but symptoms often do not reveal the cause, so tickets are passed between teams several times [1].
- "Shielding" stops a learning agent from breaking set rules, but it has been tested mainly on robots and games [2].
- "Learning to defer" passes hard cases to a person, but gives no option to gather more evidence first [3].

Gap. In the 46 sources I reviewed, these ideas appear only two at a time. No study puts all three choices in one agent for IT incidents, checks it against rules, and tests it as the evidence gets worse.

Contribution. (1) A simulator of IT incidents in which the quality of the evidence can be turned up or down. (2) An agent with the three choices and a separate rule checker. (3) A fair comparison with six simpler strategies on four measures, not accuracy alone.

Full research pipeline



What is being researched: how the three-way decision behaves under rules and poor evidence, not simply training a learning agent.

Progress to date

Done (evidence in brackets)

- Literature review: five databases searched, 46 sources reviewed, gap confirmed [Chapter 2].
- Report: Chapters 1 to 3 written [40-page draft].
- Design: simulator, agent, rule checker, comparisons and measures specified [Chapter 3].
- Partner: the Directorate of University ICT Services has agreed to support the study.

Not yet done. No code, no information collected from the Directorate and no experiments, so no results are claimed.

Next six weeks

W1: collect causes, checks and rules; build the simulator.
W2: build the rule checker and comparison strategies; Directorate staff review the simulated incidents.
W3: train the agent.
W4: compare it with the six strategies.
W5: test with worse evidence; analyse.
W6: finish the prototype, poster and results.

Prototype. The simulator and trained agent, with a simple screen that replays an incident step by step beside the comparison results.

Evaluation. The four measures above, averaged over 10 runs.

Key references

[1] Chen, Z., et al. (2020). Towards intelligent incident management: Why we need it and how we make it. *Proc. ACM ESEC/FSE*, 1487–1497. [2] Alshiekh, M., Bloem, R., Ehlers, R., Könighofer, B., Niekum, S., & Topcu, U. (2018). Safe reinforcement learning via shielding. *Proc. AAAI Conference on Artificial Intelligence*, 32(1). [3] Madras, D., Pitassi, T., & Zemel, R. (2018). Predict responsibly: Improving fairness and accuracy by learning to defer. *Advances in Neural Information Processing Systems*, 31.